Ejercicio 3.2: Imputación de datos perdidos (simple)

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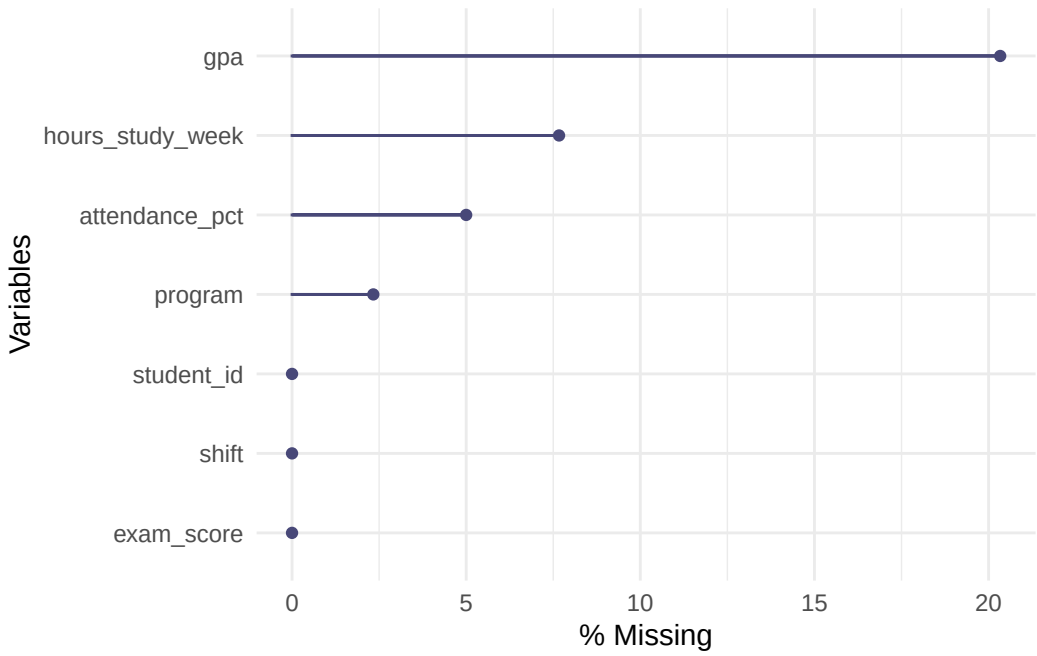
Carga de Datos y Librerías

```r
library(naniar)
library(tidyverse)
```

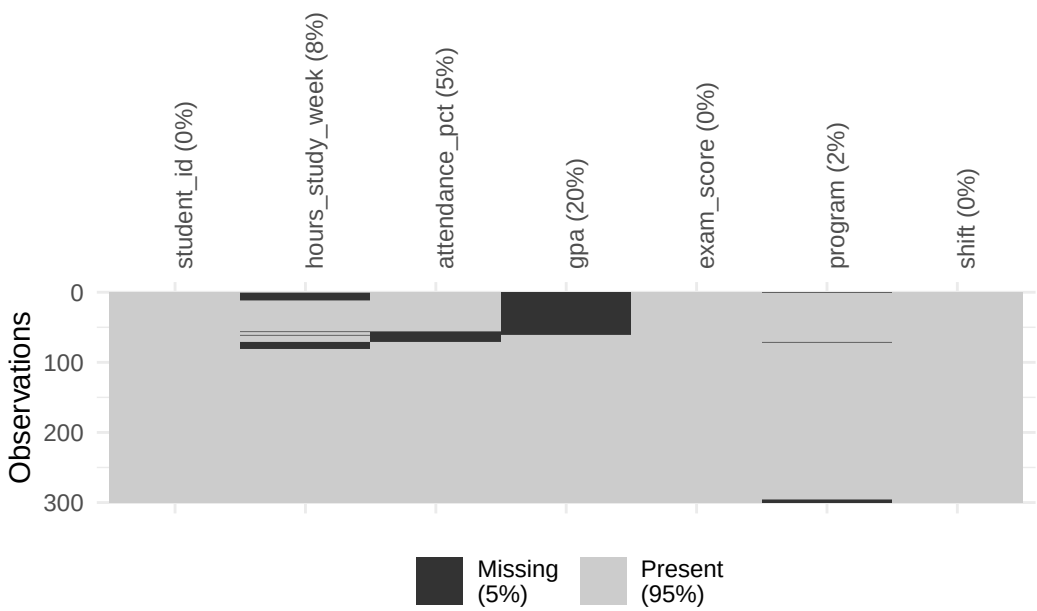
```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.0     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.2.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(rio)

data <- read.csv("students_FP.csv",
  na.strings = c("", "NA", "NaN", "NULL"),
  stringsAsFactors = TRUE
)
```

```r
gg_miss_var(data, show_pct = TRUE)
```




```
vis_miss(data,cluster = TRUE)  +
  theme(axis.text.x = element_text(angle = 90))
```



1.Imputación de las variables MCAR

La variable `program` tienen $< 5\%$ y es cualitativs, por tanto la forma más simple de imputarlas sería por la moda.

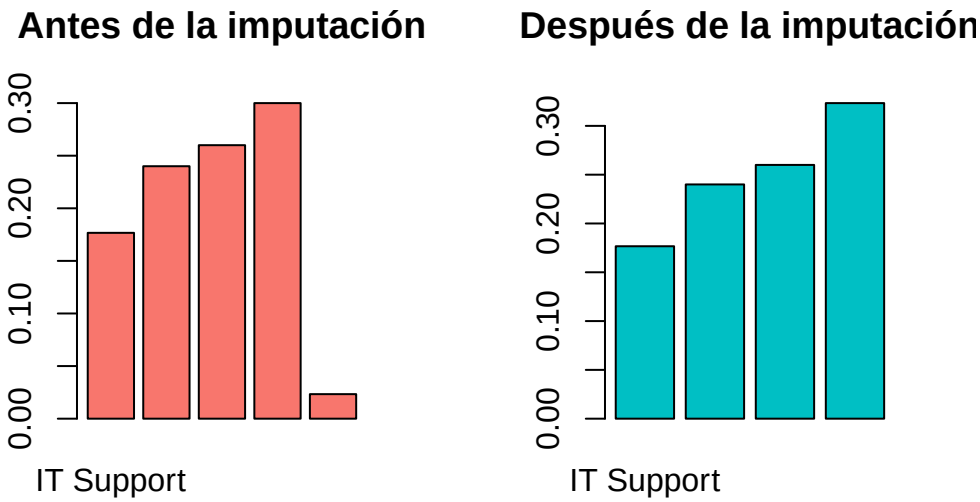
```
#Comprobar la categoría más frecuente
prop.table(table(data$program,useNA = "always"))
```

```
            IT Support Network Administration   Software Engineering
            0.17666667             0.24000000             0.26000000
       Web Development                   <NA>
            0.30000000             0.02333333
```

```
## Sustituir los NA con la categoría más frecuente
data$program_imp <-data$program
data$program_imp[is.na(data$program_imp)] <- "Web Development"

## Volvemos a comprobar
prop.table(table(data$program_imp,useNA = "always"))
```

```
            IT Support Network Administration   Software Engineering
             0.1766667              0.2400000              0.2600000
       Web Development                   <NA>
             0.3233333              0.0000000
```

```
# Graficar la distribución de la variable categórica antes y después de la imputación
par(mfrow = c(1, 2)) # Organizar las gráficas en una fila de 2 columnas
barplot(prop.table(table(data$program, useNA = "ifany")),
        main = "Antes de la imputación", col = "#F8766D" )
barplot(prop.table(table(data$program_imp)),
        main = "Después de la imputación",col = "#00BFC4")
```



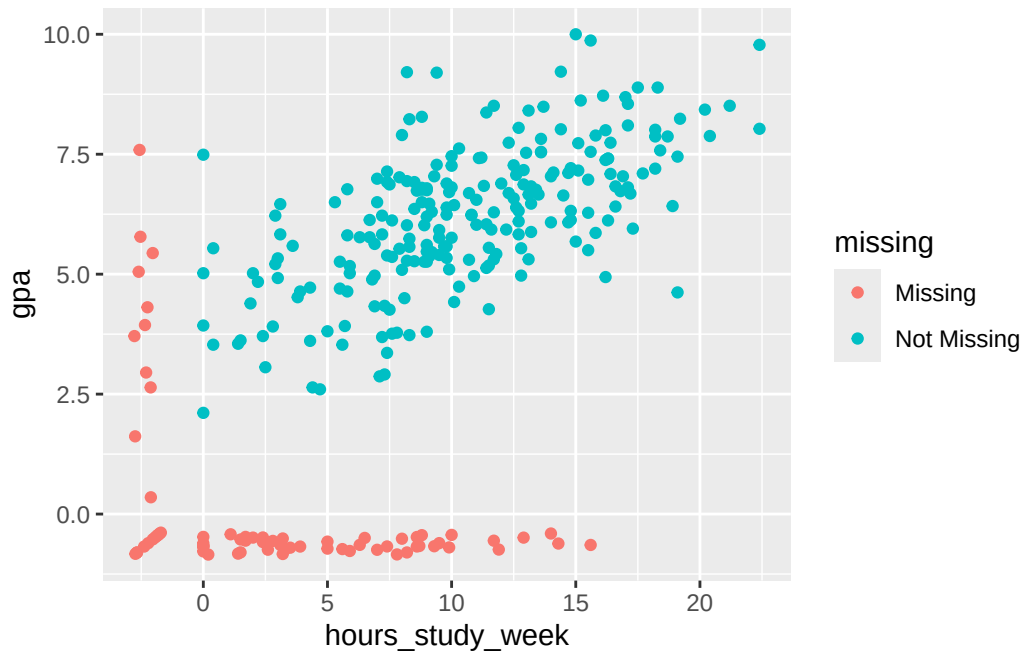
El % de datos missing de la variable program es muy bajo y al imputar por la moda, no cambia la distribución en absoluto, por tanto, lo asumiremos como una buena imputación

2.Imputación de las variables MAR/MNAR

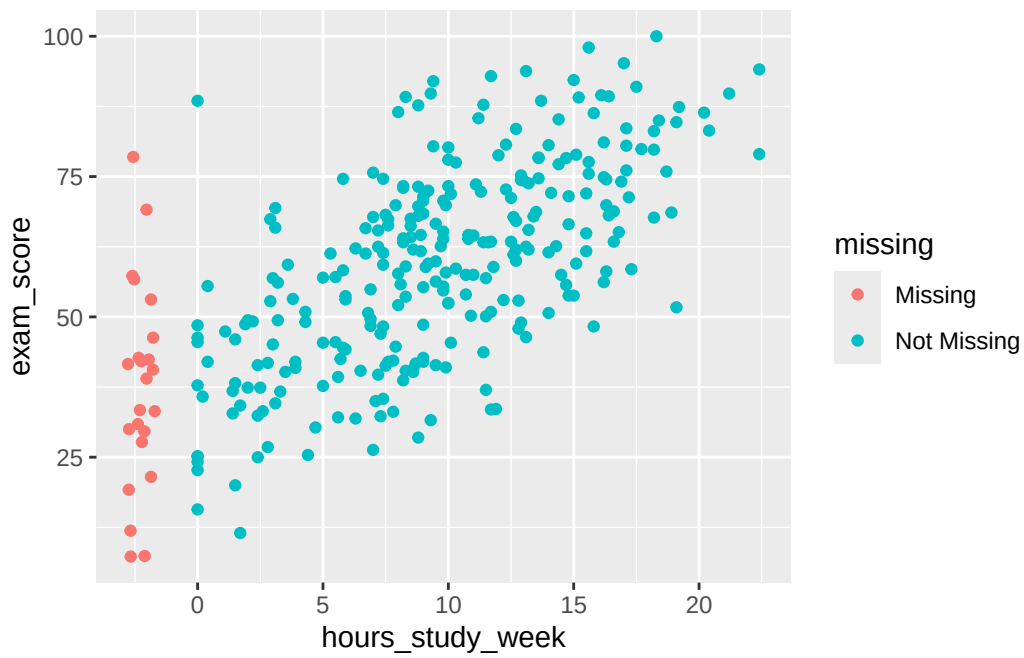
- Variable `hours_study_week`

Los NA de esta variable son MAR/MNAR y corresponden a un 8%, además la variable está asociada con gpa y exam_score:

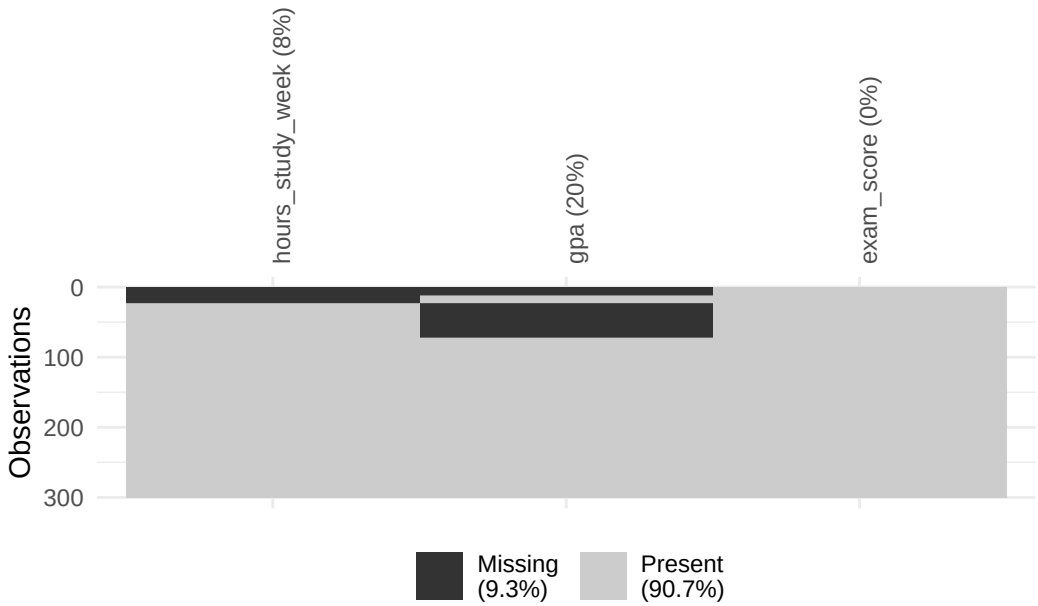
```
ggplot(data = data, aes (x = hours_study_week ,
                         y =gpa )) + geom_miss_point()
```




```
ggplot(data = data, aes (x = hours_study_week ,
                         y =exam_score )) + geom_miss_point()
```



```r
vis_miss(select(data,hours_study_week,gpa,exam_score),cluster=TRUE)  +
  theme(axis.text.x = element_text(angle = 90))
```



Como solo están completos los datos de `exam_score` para los NA de `hours_study_week`, la imputaremos mediante una regresión con esa variable:

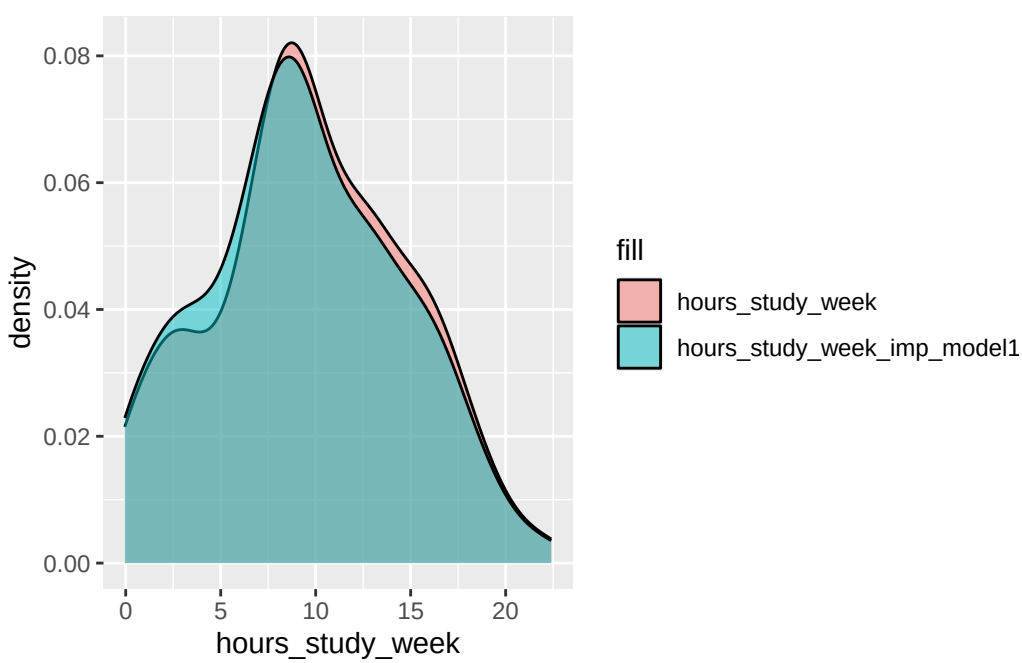
```r
#Ajustar un modelo de regresión lineal
model1 <- lm(hours_study_week ~ exam_score, data = data)

#Predecir los valores solo para las observaciones faltantes
predictions <- predict(model1,newdata = data [is.na(data$hours_study_week),])

##Crear una nueva variable imputada
data$hours_study_week_imp_model1 <- data$hours_study_week
data$hours_study_week_imp_model1[is.na(data$hours_study_week)]<- predictions

summary(model1)
```

```
Call:
lm(formula = hours_study_week ~ exam_score, data = data)

Residuals:
     Min       1Q   Median       3Q      Max
```

```
-15.0151  -2.8472   0.0736   2.6104  10.9111

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.40111    0.82399   -1.70   0.0902 .
exam_score   0.18549    0.01322   14.03   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.906 on 275 degrees of freedom
  (23 observations deleted due to missingness)
Multiple R-squared:  0.4172,    Adjusted R-squared:  0.4151
F-statistic: 196.9 on 1 and 275 DF,  p-value: < 2.2e-16
```

```r
#Hacer un gráfico para comparar las observaciones
ggplot(data, aes(x = hours_study_week, fill = "hours_study_week")) +
  geom_density(alpha = 0.5) +

geom_density(aes(x = hours_study_week_imp_model1, fill = "hours_study_week_imp_model1"), alpl
```

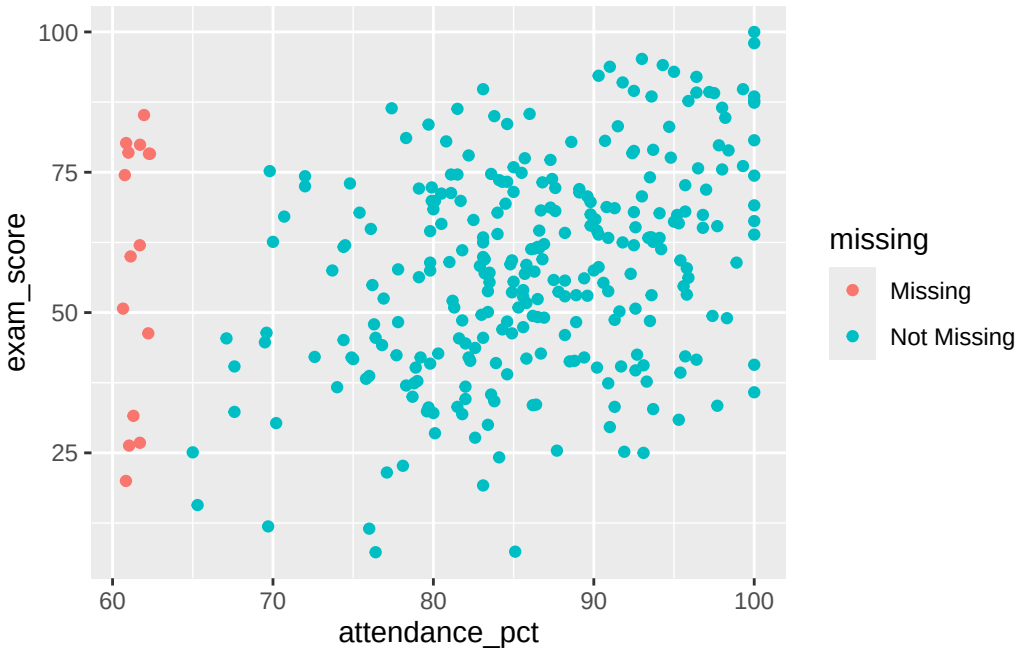
```
Warning: Removed 23 rows containing non-finite outside the scale range
(`stat_density()`).
```



- Variable `attendance_pct`

Los `NA` de `attendance_pct` son MNAR, pero con un % pequeño y además esta variable está asociada a exam_score y todos sus missing están completos

```
ggplot(data = data, aes (x = attendance_pct ,
                         y =exam_score )) + geom_miss_point()
```



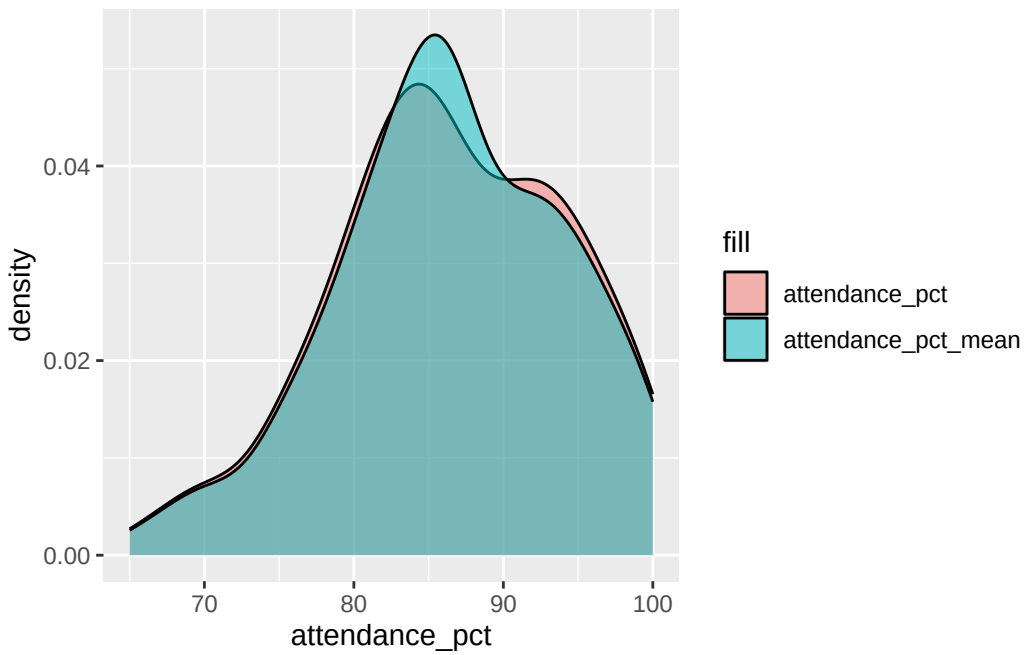
Como solo están completos los datos de `exam_score` para los NA de `attendance_pct`, la imputaremos mediante una regresión:

```
data$attendance_pct_mean<-data$attendance_pct #Nueva variable
mean(data$attendance_pct, na.rm = TRUE) #Media
```

```
[1] 86.45123
```

```
# 86.45123

data$attendance_pct_mean[is.na(data$attendance_pct_mean)] <- mean(data$attendance_pct_mean, r

##Gráfico para representar las diferencias
ggplot(data, aes(x = attendance_pct, fill = "attendance_pct")) +
  geom_density(alpha = 0.5) +
  geom_density(aes(x = attendance_pct_mean, fill = "attendance_pct_mean"), alpha = 0.5)
```

```
Warning: Removed 15 rows containing non-finite outside the scale range
(`stat_density()`).
```



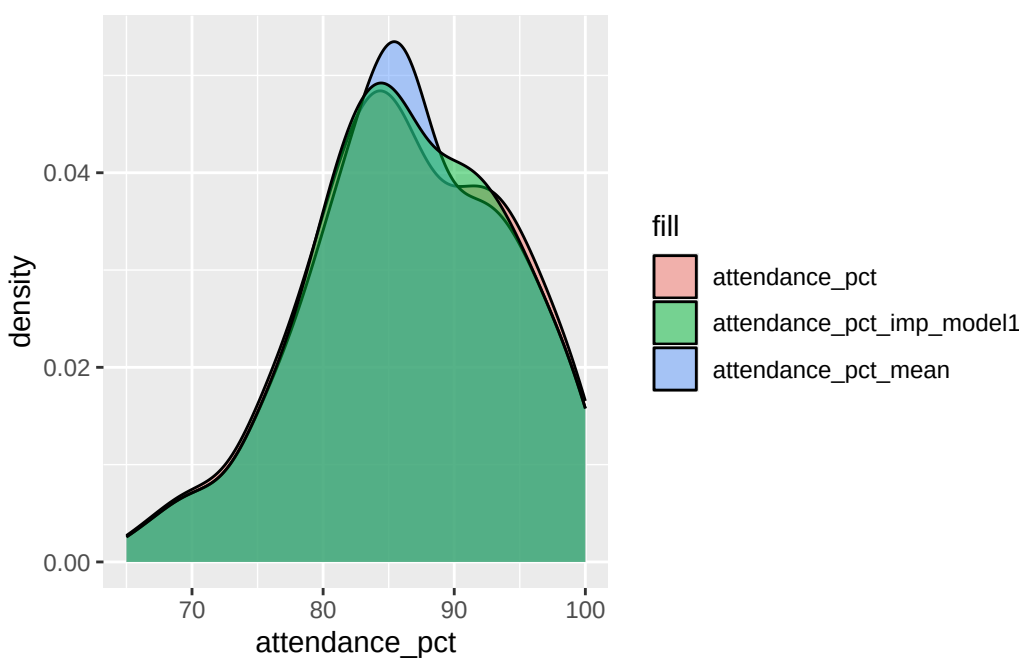

```r
#Ajustar un modelo de regresión lineal
model1 <- lm(attendance_pct ~ exam_score, data = data)
summary(model1)
```

```
Call:
lm(formula = attendance_pct ~ exam_score, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-19.4024  -4.3054  -0.0134   5.5110  17.1033 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 77.16726    1.41751  54.439  < 2e-16 ***
exam_score   0.16004    0.02328   6.874 3.98e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.269 on 283 degrees of freedom
```

```
  (15 observations deleted due to missingness)
Multiple R-squared:  0.1431,    Adjusted R-squared:  0.1401
F-statistic: 47.26 on 1 and 283 DF,  p-value: 3.977e-11
```

```r
#Predecir los valores solo para las observaciones faltantes
predictions <- predict(model1,newdata = data [is.na(data$attendance_pct),])

##Crear una nueva variable imputada
data$attendance_pct_imp_model1 <- data$attendance_pct
data$attendance_pct_imp_model1[is.na(data$attendance_pct_imp_model1)]<- predictions



#Hacer un gráfico para comparar las observaciones
ggplot(data, aes(x = attendance_pct, fill = "attendance_pct")) +
  geom_density(alpha = 0.5) +
  geom_density(aes(x = attendance_pct_mean,
                   fill = "attendance_pct_mean"), alpha = 0.5) +
  geom_density(aes(x = attendance_pct_imp_model1,
                   fill = "attendance_pct_imp_model1"), alpha = 0.5)
```

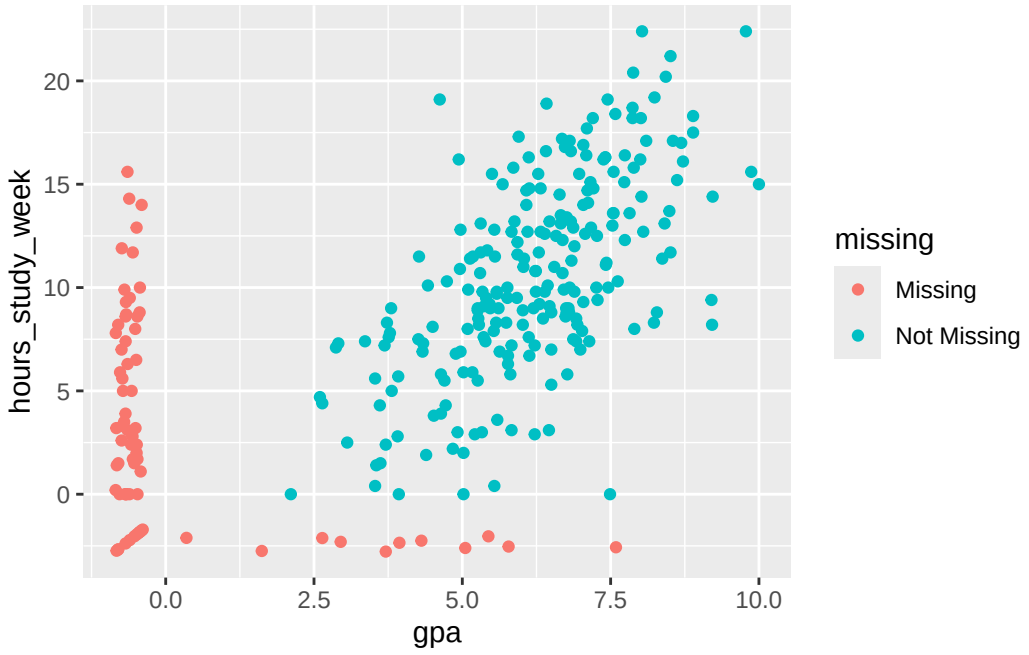
```
Warning: Removed 15 rows containing non-finite outside the scale range
(`stat_density()`).
```



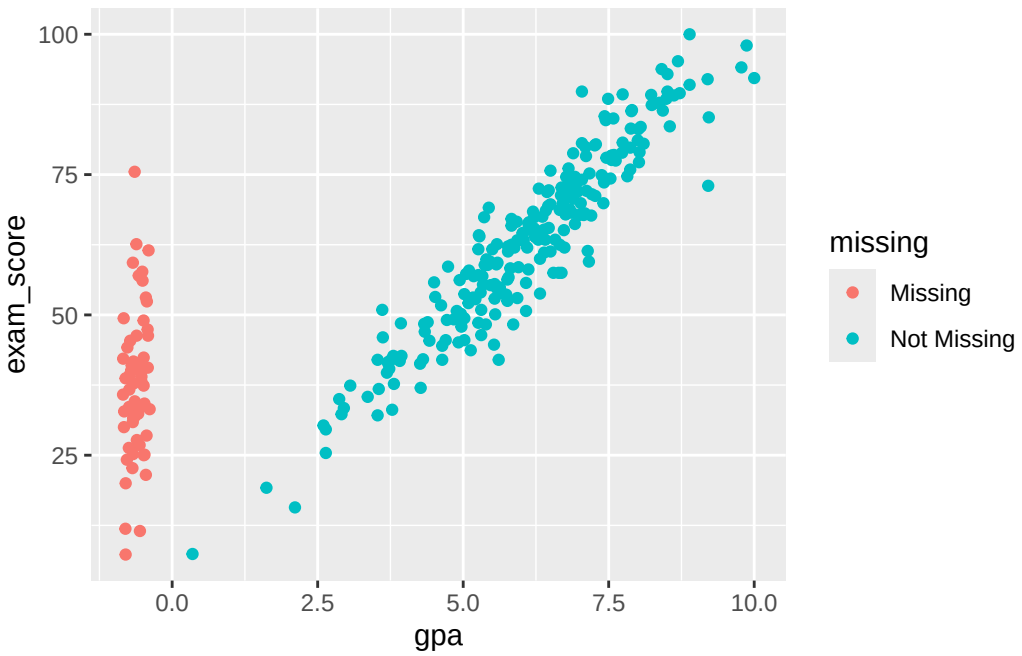
- Variable gpa

Los NA de gpa corresponden mayoritariamente a valores con datos inferiores de hours_study_week y con datos menores de exam_score.

```
ggplot(data = data, aes (x = gpa ,
                         y =hours_study_week )) + geom_miss_point()
```




```
ggplot(data = data, aes (x = gpa ,
                         y =exam_score )) + geom_miss_point()
```



En este caso, no tenemos ninguna de las dos variables asociadas como completas. Por tanto una opción es usar una de las ya imputadas. En este caso, usaremos exam_score que tiene el menor número de datos faltantes y una asociación muy clara

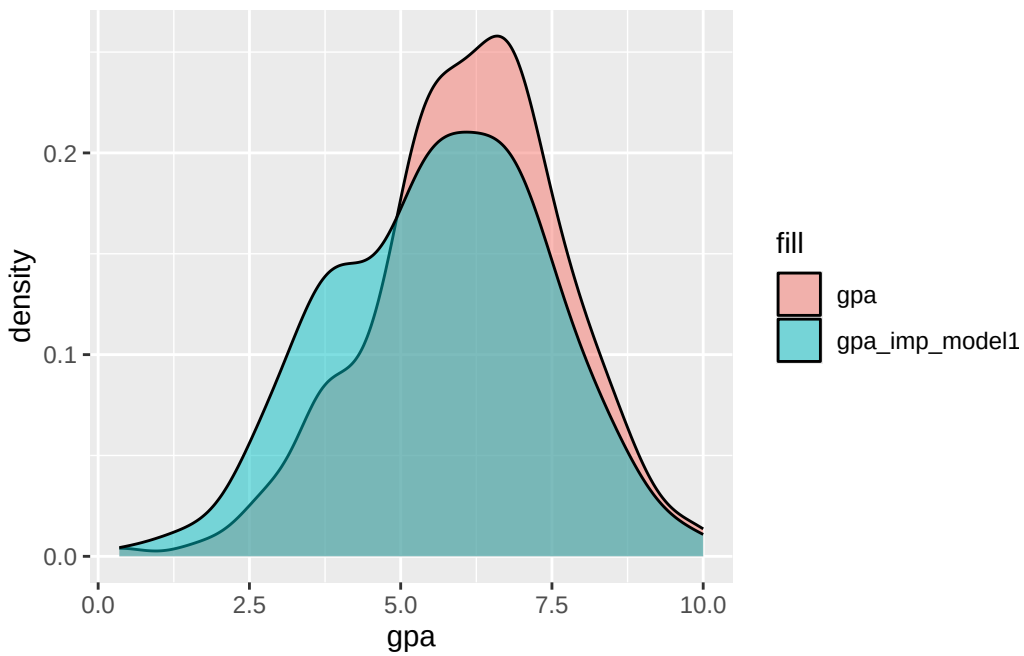
```
#Ajustar un modelo de regresión lineal
model1 <- lm(gpa ~ exam_score, data = data)
summary(model1)
```

```
Call:
lm(formula = gpa ~ exam_score, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.49257 -0.33300 -0.02257  0.28643  2.20431 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.371042   0.135401    2.74  0.00661 ** 
exam_score  0.090886   0.002075   43.81  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5256 on 237 degrees of freedom
```

```
  (61 observations deleted due to missingness)
Multiple R-squared:  0.8901,    Adjusted R-squared:  0.8896
F-statistic:  1919 on 1 and 237 DF,  p-value: < 2.2e-16
```

```r
#Predecir los valores solo para las observaciones faltantes
predictions <- predict(model1,newdata = data [is.na(data$gpa),])

##Crear una nueva variable imputada
data$gpa_imp_model1 <- data$gpa
data$gpa_imp_model1[is.na(data$gpa_imp_model1)]<- predictions

#Hacer un gráfico para comparar las observaciones
ggplot(data, aes(x = gpa, fill = "gpa")) +
  geom_density(alpha = 0.5) +
  geom_density(aes(x = gpa_imp_model1, fill = "gpa_imp_model1"),
               alpha = 0.5)
```

```
Warning: Removed 61 rows containing non-finite outside the scale range
(`stat_density()`).
```



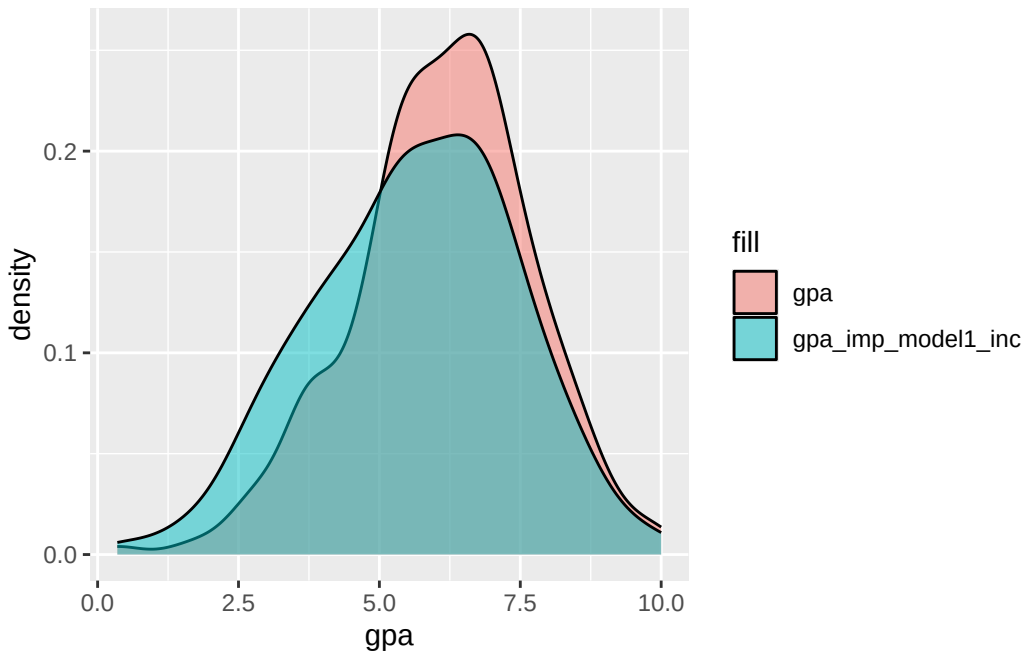
En este caso vemos que la imputación no es tan buena como en las anteriores, se puede ver si añadiendo incertidumbre mejora

```r
summary(model1) ##Cogemos el residual standard error
```

```
Call:
lm(formula = gpa ~ exam_score, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.49257 -0.33300 -0.02257  0.28643  2.20431 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.371042   0.135401    2.74  0.00661 ** 
exam_score  0.090886   0.002075   43.81  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5256 on 237 degrees of freedom
  (61 observations deleted due to missingness)
Multiple R-squared:  0.8901,	Adjusted R-squared:  0.8896 
F-statistic:  1919 on 1 and 237 DF,  p-value: < 2.2e-16
```

```r
set.seed(3)
inc<-rnorm(sum(is.na(data$gpa)), 0, sd = 0.5256)

data$gpa_imp_model1_inc <- data$gpa
data$gpa_imp_model1_inc[is.na(data$gpa)]<- predictions + inc

#Hacer un gráfico para comparar las observaciones con la media y la regresión
ggplot(data, aes(x = gpa, fill = "gpa")) +
  geom_density(alpha = 0.5) +
  geom_density(aes(x = gpa_imp_model1_inc, fill = "gpa_imp_model1_inc"),
               alpha = 0.5)
```

```
Warning: Removed 61 rows containing non-finite outside the scale range
(`stat_density()`).
```



tampoco parece muy acertado, las razones son que el % de faltantes es muy elevado y que aquí igual estamos bajo el suspuesto de MNAR y no de MAR.